

**Adventist University of Central Africa**

Master's in Big Data Analytics

# **Association Rule Learning Groceries Dataset**

*Introduction to Statistical Learning with Applications in R*

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### (a) Why these thresholds were chosen

The minimum support of **0.02** keeps the analysis focused on itemsets that appear in a meaningful share of the 9835 transactions. The minimum confidence of **0.25** removes rules that make weak predictions about their consequent. These are practical screening thresholds, not universal statistical cutoffs. Lowering either threshold would produce more rules, including rarer and potentially less stable patterns; raising them would produce a shorter list focused on the most common relationships. The thresholds should therefore be interpreted together with support, confidence, and lift rather than as evidence that every retained rule is important.

$\text{extbf}\{\text{other vegetables}, \text{yogurt}\} \Rightarrow \{\text{whole milk}\}$ , confidence = **0.5129**

### (b) Number of transactions

$\text{extbf}\{\text{other vegetables}, \text{whole milk}\} \Rightarrow \{\text{root vegetables}\}$ , lift = **2.8421** market basket.

$\text{extbf}\{\text{other vegetables}\} \Rightarrow \{\text{whole milk}\}$ , support = **0.0748** The dataset contains **169 unique grocery items**.

### (c) Ten most frequently purchased items

Table 1 and Figure 1 show the ten items with the highest support (relative purchase frequency).

Table 1: Top 10 most frequent items

| Item             | Support | Count |
|------------------|---------|-------|
| Whole milk       | 0.2555  | 2,513 |
| Other vegetables | 0.1935  | 1,903 |
| Rolls/buns       | 0.1839  | 1,809 |
| Soda             | 0.1744  | 1,715 |
| Yogurt           | 0.1395  | 1,372 |
| Bottled water    | 0.1105  | 1,087 |
| Root vegetables  | 0.1090  | 1,072 |
| Tropical fruit   | 0.1049  | 1,032 |
| Shopping bags    | 0.0985  | 969   |
| Sausage          | 0.0940  | 924   |

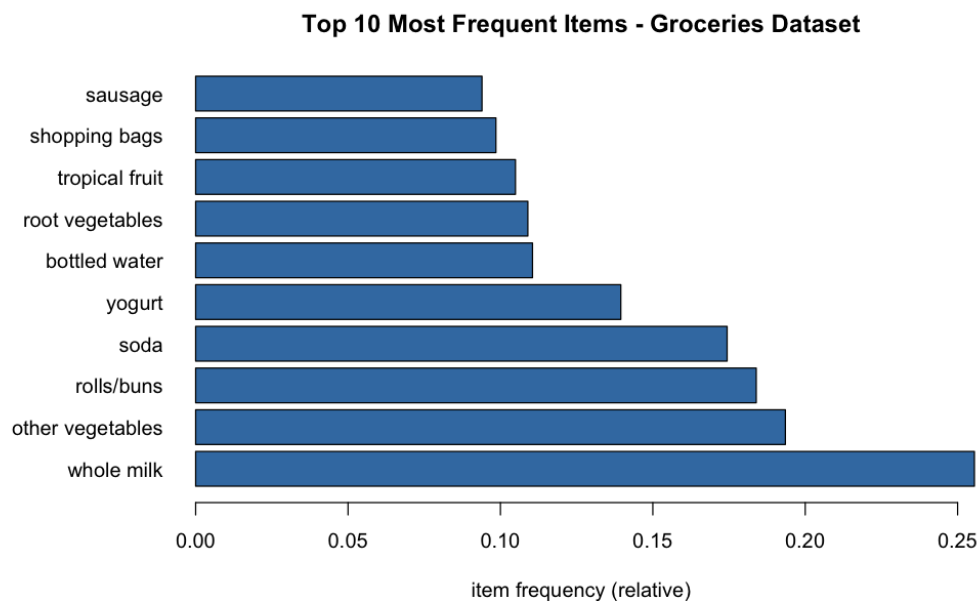


Figure 1: Top 10 most frequent items by support

Whole milk and other vegetables are, by a clear margin, the two staples of this store's customer base – unsurprising, since both are everyday perishables bought on nearly every shopping trip.

## Q1. Frequent Itemsets

### (a) At least 10 frequent itemsets (two or more products)

Using a minimum support threshold of **0.02** (i.e. the itemset appears in at least 2% of all transactions) and a minimum itemset size of 2, the analysis identified **63 frequent itemsets** (61 pairs and 2 triples). Table 2 lists the top 10 by support.

Table 2: Top 10 frequent itemsets by support

| Itemset                             | Support | Count |
|-------------------------------------|---------|-------|
| {other vegetables, whole milk}      | 0.0748  | 736   |
| {rolls/buns, whole milk}            | 0.0566  | 557   |
| {whole milk, yogurt}                | 0.0560  | 551   |
| {root vegetables, whole milk}       | 0.0489  | 481   |
| {other vegetables, root vegetables} | 0.0474  | 466   |
| {other vegetables, yogurt}          | 0.0434  | 427   |
| {other vegetables, rolls/buns}      | 0.0426  | 419   |
| {tropical fruit, whole milk}        | 0.0423  | 416   |
| {soda, whole milk}                  | 0.0401  | 394   |
| {rolls/buns, soda}                  | 0.0383  | 377   |

### (b) Support values

Support values are reported directly in Table 2 (column *Support*), ranging from 0.0383 to 0.0748 for the top 10 itemsets.

### (c) Highest-support itemset

{other vegetables, whole milk} has the highest support at **0.0748** (736 of 9,835 transactions) – almost twice the support of the next-highest itemset.

### (d) What support tells us about purchasing behavior

Support measures how common an itemset is across *all* transactions, regardless of what else is happening in the basket. A support of 0.0748 for {other vegetables, whole milk} means that roughly 1 in 13 shopping trips includes both items together. High-support itemsets identify the combinations that matter at scale – store layout, staffing, and inventory decisions built around them affect a large share of all customers, whereas a rule with strong confidence but tiny support only describes a small, possibly niche, slice of shoppers.

## Q2. Association Rules

### Measures used

Three measures are used throughout this report, for a rule  $A \Rightarrow B$  and an itemset  $X$  (with  $T$  the set of all transactions):

$$\text{support}(X) = \frac{|\{t \in T : X \subseteq t\}|}{|T|} \quad \text{confidence}(A \Rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)} \quad \text{lift}(A \Rightarrow B) = \frac{\text{confidence}(A \Rightarrow B)}{\text{support}(B)}$$

### (a) At least 10 association rules

Rules were generated from the frequent itemsets above using a minimum confidence of **0.25**, yielding **49rules** in total. Table 3 lists the top 12, sorted by lift.

Table 3: Association rules (sorted by lift)

| Antecedent                   | Consequent       | Support | Confidence | Lift   |
|------------------------------|------------------|---------|------------|--------|
| other vegetables, whole milk | root vegetables  | 0.0232  | 0.3098     | 2.8421 |
| pip fruit                    | tropical fruit   | 0.0204  | 0.2702     | 2.5746 |
| root vegetables, whole milk  | other vegetables | 0.0232  | 0.4740     | 2.4498 |
| root vegetables              | other vegetables | 0.0474  | 0.4347     | 2.2466 |
| other vegetables, whole milk | yogurt           | 0.0223  | 0.2976     | 2.1330 |
| whipped/sour cream           | other vegetables | 0.0289  | 0.4028     | 2.0819 |
| whipped/sour cream           | yogurt           | 0.0207  | 0.2894     | 2.0743 |
| whole milk, yogurt           | other vegetables | 0.0223  | 0.3975     | 2.0541 |
| other vegetables, yogurt     | whole milk       | 0.0223  | 0.5129     | 2.0072 |
| tropical fruit               | yogurt           | 0.0293  | 0.2791     | 2.0005 |
| butter                       | whole milk       | 0.0276  | 0.4972     | 1.9461 |
| pork                         | other vegetables | 0.0217  | 0.3757     | 1.9415 |

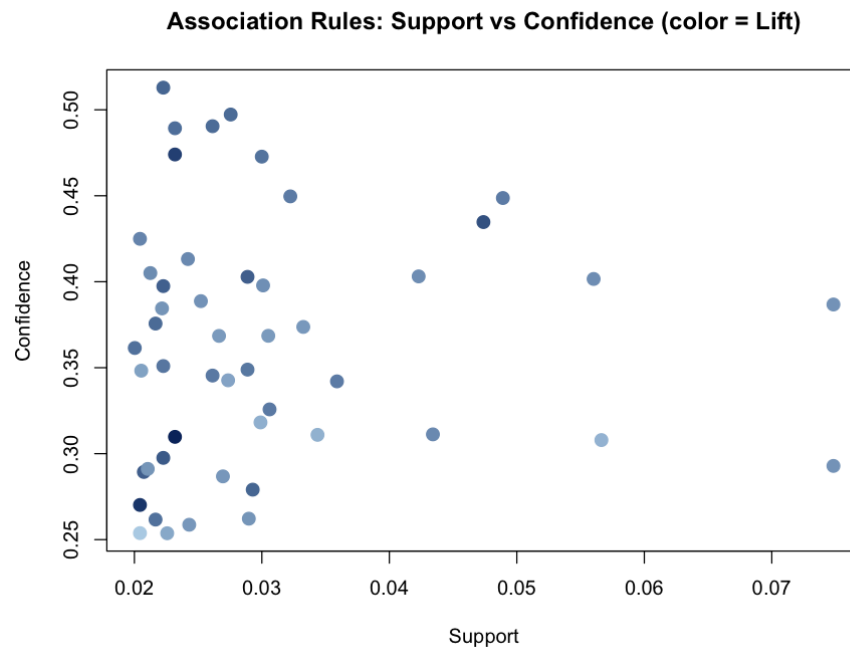


Figure 2: All 49 rules: support vs. confidence, coloured by lift (darker = higher lift)

### Q3. Understanding the Measures

Three rules from Table 3 are examined below.

#### (a) Support

- **{other vegetables, whole milk} → {root vegetables}**:

$$\text{support} = \frac{|\{t : \{\text{o.veg, milk, root veg}\} \subseteq t\}|}{9,835} = \frac{228}{9,835} = 0.0232$$

this exact combination of all three items occurs in 2.32% of all transactions.

- **{root vegetables} → {other vegetables}**: support = 0.0474 – both items together appear in 4.74% of transactions.
- **{butter} → {whole milk}**: support = 0.0276 – both items appear together in 2.76% of transactions.

#### (b) Confidence

- **{other vegetables, whole milk} → {root vegetables}**:

$$\text{confidence} = \frac{\text{support}(A \cup B)}{\text{support}(A)} = \frac{0.0232}{0.0748} = 0.3098$$

of the customers who bought other vegetables *and* whole milk, 30.98% also bought root vegetables.

- **{root vegetables} → {other vegetables}**: confidence = 0.4347 – 43.47% of customers who bought root vegetables also bought other vegetables.
- **{butter} → {whole milk}**: confidence = 0.4972 – almost half (49.72%) of butter buyers also bought whole milk.

#### (c) Lift

- **{other vegetables, whole milk} → {root vegetables}**:

$$\text{lift} = \frac{\text{confidence}(A \Rightarrow B)}{\text{support}(B)} = \frac{0.3098}{0.1090} = 2.8421$$

customers who buy other vegetables and whole milk are 2.84 times more likely to also buy root vegetables than a random customer.

- **{root vegetables} → {other vegetables}**: lift = 2.25 – buying root vegetables makes buying other vegetables 2.25 times more likely than chance.
- **{butter} → {whole milk}**: lift = 1.95 – buying butter makes buying whole milk about 1.95 times more likely than chance, despite whole milk already being the most popular item in the store.

## Q4. Find Interesting Rules

### (a) Highest confidence

$\{\text{other vegetables, yogurt}\} \rightarrow \{\text{whole milk}\}$ , confidence = 0.5129 (support = 0.0223, lift = 2.01).

### (b) Highest lift

$\{\text{other vegetables, whole milk}\} \rightarrow \{\text{root vegetables}\}$ , lift = 2.84 (support = 0.0232, confidence = 0.31).

### (c) Highest support

$\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$ , support = 0.0748 (confidence = 0.3868, lift = 1.51).

### (d) Are these the same rule?

No – all three are different rules. This is expected: support rewards *popularity* (the highest-support rule involves the two most common items in the store, so it occurs often even though the association is only moderately strong – lift 1.51 is the lowest of the three). Confidence rewards *reliability* of the antecedent predicting the consequent, which is why the highest-confidence rule points at whole milk, the single most-bought item overall – it's an easy consequent to satisfy. Lift rewards *genuine association strength* after removing the effect of overall popularity, which is why the highest-lift rule involves root vegetables, a less common item whose purchase is still strongly tied to a specific combination of other items. A rule can score well on one measure and only moderately on the others, since each measure is answering a different question.

## Q5. Interpret a Rule

Selected rule: **{other vegetables, whole milk} → {root vegetables}** (support = 0.0232, confidence = 0.31, lift = 2.84).

**In simple language:** Customers who purchase other vegetables and whole milk in the same trip also tend to purchase root vegetables in about 31% of those trips – noticeably more often than root vegetables are bought overall (support 0.109), which is exactly what the lift of 2.84 captures.

**Strong or weak association?** This rule represents a **moderately strong** association. The lift of 2.84 is comfortably above 1, meaning the three items co-occur far more than chance would predict, and 31% confidence means nearly a third of the relevant customers follow the pattern. It is not an overwhelming rule (confidence is well under 50%), but combined with a lift close to 3, it is one of the more meaningful relationships found in this dataset – plausibly reflecting a shared ”cooking a vegetable-based meal” shopping occasion.

## Q6. Business Application

Imagine acting as the supermarket manager, using the rules above.

### (a) Two products to place close together

**Root vegetables and other vegetables.** Their rule ( $\{\text{root vegetables}\} \rightarrow \{\text{other vegetables}\}$ , lift = 2.25) shows a strong, high-confidence association (43.5%), and both are fresh produce that is naturally merchandised in the same aisle already – placing them adjacent (or cross-merchandising root vegetables at the front of the vegetable section) would shorten the trip for a large share of customers.

### (b) One possible product bundle

A **”vegetable basket” bundle of other vegetables + whole milk + root vegetables** (the highest-lift itemset, lift 2.84) – e.g. a discounted ”dinner veg box” combining these three – targets an already-existing shopping pattern rather than trying to create a new one.

### (c) Using association rules for recommendations

At checkout (in-store) or in an online cart, the system could flag: ”Customers who bought other vegetables and whole milk also bought root vegetables (frequently)” and offer a small discount or a reminder nudge – a ”you might also need” prompt driven directly by the rule  $\{\text{other vegetables, whole milk}\} \rightarrow \{\text{root vegetables}\}$ , rather than a generic or manually curated suggestion.

## Q7. Understanding Association

### (a) Does an association rule prove causation?

No. An association rule only describes **co-occurrence** in the transaction data – that customers who bought  $A$  frequently also bought  $B$ . It says nothing about *why*:  $A$  might not cause anyone to buy  $B$  at all. Both could be driven by a common cause (e.g. both are ingredients for the same popular recipe), by seasonality, by store layout placing them near each other, or by pure coincidence in a large dataset. Confirming causation would require a controlled experiment (e.g. moving product placement and measuring the effect), not just mining historical baskets.

### (b) High confidence but low lift

This happens when the consequent is already very common on its own. Confidence only asks "given the antecedent, how often does the consequent follow?" – if the consequent (say, whole milk, support 0.2555) is bought in a quarter of *all* transactions regardless of anything else, then almost any antecedent will show reasonably high confidence for predicting it, purely because it's hard to miss. Lift corrects for this by dividing by the consequent's baseline popularity, so a rule can have confidence well above the dataset average while still having lift close to 1 – meaning the antecedent barely changes the odds of the consequent compared to knowing nothing at all.

### (c) What a lift greater than 1 indicates

Lift  $> 1$  indicates a **positive association**: the two itemsets occur together more often than would be expected if they were purchased independently. The further above 1, the stronger the association (lift = 1 means no association / statistical independence, and lift  $< 1$  indicates a negative association, where buying one item makes the other *less* likely).

## Q8. Limitations and Responsible Use

The results describe this transaction sample under the selected thresholds; they do not establish that the rules will hold in every store, season, or customer segment. Rare rules are especially sensitive to a small number of baskets, and repeated searches over many possible rules increase the chance of finding apparently impressive patterns by chance. The analysis also treats each transaction as a basket and does not include price, quantity, customer identity, time, promotion, or store-location information. These omitted variables could explain part of the observed associations.

For a production recommendation system, candidate rules should therefore be tested on a later holdout period or a different store before deployment. A/B testing product placement or recommendation messages would be needed to estimate whether a proposed intervention actually changes purchasing behaviour.

### (a) Choosing a rule for a business decision

The appropriate metric depends on the decision. High-support rules are useful for broad inventory and store-layout decisions because they affect many baskets. High-confidence rules are useful when the business wants a reliable follow-up recommendation after an antecedent has already been observed. High-lift rules are useful for targeted cross-selling because they identify combinations that are unusually strong relative to the consequent's baseline popularity. In all three cases, the metric should be considered alongside support: a very high lift based on only a handful of transactions may not be stable enough to act on.